

Satellite-based power outage prediction for Mexico

Calibrated probabilistic forecasting with nighttime lights

Armande Aboudrar-Méda, Giacomo Falchetta
CMCC

Executive Summary

Update: shifted from Top-K / Learning-to-Rank to **calibrated probabilistic forecasting** as primary pipeline.

New framing: “What is the probability of a ≥ 3 h outage tonight?” — rather than “which K municipalities should we inspect?”

What changed since last meeting:

1. Replaced LTR with **binary XGBoost + Platt calibration** as primary model
2. Systematic **7-model ablation** to identify which features matter
3. **Cause-specific evaluation** (Environmental vs Technical)
4. New robustness tests: Texas sensitivity, spatial holdout
5. LTR retained as **secondary benchmark** for fixed-budget comparison

Headline: ROC-AUC **0.879**, PR-AUC **0.056** (14× random), Precision **12.5%** at ~8 alerts/day (25× base rate). Environmental outages significantly more predictable than Technical ($p < 0.05$).

Why Shift from Top-K/LTR to Probability + Calibration?

What we had (LTR):

- ▶ XGBoost `rank:pairwise` — optimises within-day ranking
- ▶ Each day, pick **top-K** municipalities
- ▶ Assumes **fixed daily alert budget**
- ▶ Scores are ordinal, **not probabilities**

When LTR is the right tool:

- ▶ A utility can dispatch exactly K inspection teams/day
- ▶ You only care about relative ordering
- ▶ Fixed-K is a hard operational constraint

Why we shifted (literature insight):

When operations **do not impose a fixed K**, a calibrated probability is the better primary model because it supports:

- ▶ **Flexible thresholds** — some days 5 alerts, others 20
- ▶ **Interpretable risk** — “5% chance tonight” vs “ranked 7th”
- ▶ **Decision curves** — optimal threshold from cost-benefit analysis
- ▶ **Downstream models** — P(outage) as input to count/duration models

Key insight from literature: Good ranking (high ROC-AUC) $\neq$ good probabilities. Post-hoc calibration is needed to turn scores into honest forecasts.

Two Approaches, Two Use Cases

	Calibrated Binary (Primary)	LTR / Top-K (Secondary)
Question	“What is P(outage tonight)?”	“Which K municipalities are most at risk?”
Assumption	Flexible alert budget	Fixed daily budget
Output	Calibrated probability [0, 1]	Ranking score (no probability meaning)
Threshold	Policy parameter (alerts/day cap, precision floor)	Fixed K
Evaluation	PR-AUC, calibration metrics, decision curves	Recall@K, NDCG, Lift@K
Strengths	Flexible, interpretable, composable	Directly optimises within-day ranking

Bottom line: The two approaches are complementary. Calibrated binary is primary (more general); LTR is the benchmark under fixed-budget assumptions.

Data Foundation (Recap)

Panel: One row per municipality × day

Data Source	Role
NASA Black Marble VNP46A2	Core signal (daily NTL at 500m)
GHS-BUILT-S (EU JRC)	Built-up weighting
ERA5 weather	Complementary signal
GADM v4.1	Municipal boundaries
CFE outage records	Ground truth labels

Dimensions:

- ▶ 2,457 municipalities × 1,817 days
- ▶ **4,464,369 rows**
- ▶ Prevalence: **~0.5%** (rare event)

Feature architecture (82 features):

Family	Count	Examples
NTL lags	~12	ntl_sum_built_lag1/7/14
NTL rolling	~8	roll7_mean , diff1
NTL anomaly	~4	z-scores, drop flags
Spatial neighbors	~6	nb_mean_ntl_sum_built_lag1
Weather (lagged)	~16	atm_lag1 , rain_lag7
Weather (same-day)	~16	Nowcast only
Calendar	~8	DOY sin/cos, holidays
Quality	~4	Coverage indicators

Temporal Split + Model Configuration

Strict temporal split (no leakage):

Split	Period	Rows	Prevalence
Train	2017-01 to 2019-12	2,683,044	0.59%
Validation	2020-01 to 2020-06	447,174	0.38%
Test	2020-07 to 2021-12	1,334,151	0.40%

Validation further split 50/50:

- ▶ Jan–Mar 2020 → fit calibration parameters
- ▶ Apr–Jun 2020 → select operating thresholds

This **strict calibration-threshold separation** prevents double-dipping.

Model configuration:

Parameter	Value
Algorithm	XGBoost (gradient-boosted trees)
Trees	500
Max depth	6
Learning rate	0.05
Subsampling	80% rows, 80% features
Class imbalance	<code>scale_pos_weight</code> = 167.2
Objective	<code>binary:logistic</code>

`scale_pos_weight`: tells the model that missing a positive is 167× worse than a false alarm, forcing it to learn outage patterns despite 0.5% prevalence.

How the Pipeline Works (Step by Step)

Step 1 — Feature engineering

For each municipality-night: NTL lags, rolling stats, anomaly z-scores, spatial neighbors, weather, calendar. **Forecast mode:** only $t-1$ features (no same-day data).

Step 2 — Train XGBoost classifier

Binary classification with `scale_pos_weight`. Trained on 2017-2019 only. Outputs scores $\in [0, 1]$.

Step 3 — Raw output $\neq$ probabilities

XGBoost's scores are useful for *ranking* but **severely miscalibrated** as probabilities. When it says “10% risk,” the true rate may be 0.3% or 25%.

Step 4 — Post-hoc calibration

Transform raw scores into honest probabilities:

Method	Parameters	Role
Platt	2 (logistic fit)	Default — stable
Beta	3 (Beta transform)	Sensitivity check
Isotonic	Non-parametric	Diagnostic only

Step 5 — Threshold = policy decision

“Alert if $P > T$.” The choice of T is operational:

- ▶ Cap 10 alerts/day $\rightarrow T = 0.110$
- ▶ Precision floor $\geq 10\%$ $\rightarrow T = 0.079$
- ▶ Maximise F1 $\rightarrow T = 0.057$

Step 6 — Evaluate on held-out test period

All metrics on Jul 2020 – Dec 2021 (never seen).

Understanding the Metrics

Discrimination: can it tell outages apart?

ROC-AUC (0.879): Pick one outage-night and one non-outage-night randomly — the model ranks the outage higher **87.9% of the time**. (Random = 0.5)

PR-AUC (0.056): How well does it retrieve the rare positives? Random = 0.004 (prevalence). Ours = **14× random**. Looks “low” but is strong for a 0.5% event.

ROC-AUC can be misleadingly high with rare events; PR-AUC is the more honest metric here.

Calibration: are the probabilities honest?

Metric	Raw	Platt	Meaning
Log-loss	0.30	0.02	93% improvement
Brier	0.09	0.004	Near theoretical min
ECE	0.20	0.001	Essentially perfect

After Platt calibration: when the model says “5% risk,” ~5% of those municipality-nights actually have outages.

Operational: what happens when deployed?

Precision (12.5%): 1 in 8 alerts is true → **25× concentration** vs random.

Recall (9.7%): catches ~1 in 10 outages at 8 alerts/day.

Main Results: Strict Forecast Benchmark

Run: 20260223_125430_forecast_strict | Mode: Forecast (strict t-1 only)

Discrimination

Metric	Validation	Test
ROC-AUC	0.886	0.879
PR-AUC	0.047	0.056

Small val→test gap (0.007) confirms no overfitting. Test PR-AUC is higher (more seasonal variation in test period).

Calibration comparison

Method	Log-loss	Brier	ECE
Raw	0.300	0.093	0.196
Platt	0.021	0.004	0.001
Beta	0.021	0.004	0.001
Isotonic	0.032	0.004	0.001

Operating points (Platt-calibrated)

Policy	Threshold	Prec.	Recall	Alerts/day
Cap 10/day	0.110	12.5%	9.7%	7.6
Prec ≥ 10%	0.079	9.8%	16.3%	16.4
Cap 20/day	0.077	9.7%	17.0%	17.3
Max F1	0.057	7.6%	23.6%	30.5

Default policy (cap 10/day): Precision 12.5% [11.5%, 13.6%], Recall 9.7% [8.9%, 10.5%].

The threshold is a **policy knob**: stricter → fewer alerts, higher precision; looser → more catches, more false alarms. No single “correct” threshold.

Top Feature Importances

Top features by XGBoost gain:

Rank	Feature	Gain	Meaning
1	ntl_sum_built_lag1	12.6%	Brightness last night
2	ntl_sum_built_lag7	9.1%	Brightness 1 week ago
3	ntl_sum_built_lag14	7.2%	Brightness 2 weeks ago
4	ntl_sum_built_lag2	5.1%	Brightness 2 nights ago
5	rain	4.6%	Rainfall
6	nb_mean_ntl_sum_built_lag1	3.6%	Neighbors' brightness
7	max_dew	3.4%	Dew point
8	min_temp	3.2%	Min temperature

Interpretation:

- ▶ **Top 4 features are NTL lags** — brightness dynamics over 1–14 days are the core signal
- ▶ **Weather** (rain, dew, temp) complements NTL
- ▶ **Spatial neighbors** (#6) — regional brightness carries information beyond individual municipalities
- ▶ Built-weighted > raw NTL throughout

Key: The model learns that outages follow persistent brightness drops + adverse weather conditions. NTL history is the dominant information source.

Ablation Study: What Features Matter?

Batch: 20260302_122455 — 7 ablations, all strict forecast mode.

Ablation	Features included	# Feat.	ROC-AUC	PR-AUC	Recall
ntl_plus_weather	NTL + weather + calendar	62	0.872	0.052	18.0%
full_model	Everything	66	0.877	0.053	17.3%
ntl_only	NTL features only	33	0.845	0.042	9.9%
ntl_plus_spatial	NTL + neighbors	30	0.845	0.042	9.1%
ntl_core	Basic NTL lags only	26	0.835	0.033	5.5%
ntl_plus_drops	NTL + drop flags	28	0.834	0.033	5.0%
weather_lagged_only	Weather + calendar (no NTL)	47	0.804	0.035	4.1%

Takeaways: (1) NTL alone reaches ~96% of full model discrimination. (2) Weather alone is weak but **doubles recall** when added to NTL (9.9% → 18.0%). (3) Spatial and drop features add negligibly. (4) **ntl_plus_weather is the operational winner.**

Ablation: Interpretation

1. NTL is the core signal

Even the simplest NTL lags (26 features) achieve ROC-AUC 0.835. This **validates the fundamental premise**: satellite nighttime lights carry substantial information about outage risk.

2. Weather alone is weak but complementary

Weather-only: ROC-AUC 0.804, Recall 4.1%. But NTL + weather nearly doubles recall. Weather helps distinguish weather-driven dimming from other causes of brightness variation.

3. Spatial neighbors add negligibly

Likely because NTL lags already encode spatial correlation (neighboring municipalities dim together during storms).

4. Anomaly/drop features add little in forecast mode

These features detect same-day brightness anomalies; their lagged versions carry little marginal information above standard lags.

Implication: The feature story is clean — NTL + weather is the dominant combination. No need for complex anomaly detection or embeddings.

Nowcast vs Forecast

Forecast (primary): Only $t-1$ features — what you'd use in a real operational setting (satellite hasn't overflown yet).

Nowcast (secondary): Also uses same-day NTL — useful for detection (“did an outage happen tonight?”).

Mode	Features	ROC-AUC	PR-AUC	Recall
Forecast	66	0.877	0.053	17.3%
Nowcast	83	0.891	0.058	18.3%

Finding: Same-day NTL provides only a **modest uplift** (+1.4pp ROC-AUC, +1pp recall).

The predictive signal is mostly in the *history* of brightness, not tonight's observation.

Implication for deployment: We can issue forecasts *before the night begins* without losing much accuracy. The nowcast serves as an **upper bound** on achievable performance.

Robustness: Texas Sensitivity

Concern: In February 2021, Storm Uri caused catastrophic outages in northern Mexico. Does this extreme event drive our test results?

Test: Re-evaluate excluding Feb 15–17, 2021 (7,371 rows, 0.55% of test set).

Model	Full Test	Excl. Texas	Change
ntl_plus_weather	0.872	0.879	+0.6pp
full_model	0.877	0.883	+0.6pp

Result: The model performs **slightly better** without the Texas storm. The extreme event adds noise (unprecedented conditions), not signal.

Main conclusions and model rankings are **unaffected**.

Robustness: Spatial Holdout (Southern Mexico)

Design: Train on all Mexico *except* Guerrero, Oaxaca, Chiapas. Test on those three states only.

Why these states? Most rural, poorest, different grid infrastructure, hurricane-prone, ~417K test rows.

Metric	Full Test	South Only	Retention
ROC-AUC	0.872	0.859	98.5%
PR-AUC	0.052	0.016	31.3%
Recall	18.0%	1.8%	9.7%
Prevalence	0.40%	0.23%	—

Interpretation (nuanced):

- ▶ **Ranking transfers well** (98.5% ROC-AUC retention) — the model can order risk in unseen geography.
- ▶ **Calibrated probabilities do not transfer** — PR-AUC and recall drop substantially due to:
 - ▶ Lower prevalence in the South (0.23% vs 0.40%)
 - ▶ Calibration mismatch (threshold tuned nationally)
 - ▶ Genuinely harder prediction (rural, sparse NTL)
 - ▶ Distribution shift (unseen municipality baselines)

Takeaway: In deployment, one would need to **re-calibrate** for new regions. This is consistent with the remote sensing generalization literature.

Cause-Specific Analysis: Motivation

Why study causes?

Different outage causes have different signatures:

- ▶ **Environmental** (storms, cold, trees, earthquakes) — should be more predictable from NTL + weather (same weather that causes outage also dims lights)
- ▶ **Technical** (equipment failure, human error, animals) — driven by infrastructure state, less visible from space
- ▶ **Planned** (maintenance) — follows utility schedules

How causes enter the panel:

- ▶ 85.8% of positive muni-nights: single outage → unambiguous
- ▶ 12.5%: multiple outages, same cause → unambiguous
- ▶ **1.7%**: mixed causes → assign most frequent

Evaluation design:

The model is **NOT retrained**. We use the same Platt-calibrated scores and ask: does the model's risk score separate one cause type from “no outage” better or worse?

One-vs-rest: For each cause C:

- ▶ $y = 1$ if (outage AND cause == C)
- ▶ $y = 0$ otherwise (incl. other-cause outages)

This is conservative — tests whether the model can specifically detect C-type outages.

Cause distribution (test period):

Cause	Positive nights	Share
Environmental	4,150	77.8%
Technical	1,055	19.8%
Planned	98	1.8%

Cause	Positive nights	Share
Other	31	0.6%

Cause-Specific Results

Performance by cause (Platt, cap 10/day):

Cause	Support	ROC-AUC	PR-AUC	Prec.	Recall
Environmental	4,150	0.883	0.041	9.8%	7.7%
Technical	1,055	0.819	0.013	3.1%	9.5%
Planned	98	0.889	0.001	0.2%	5.1%
Other	31	0.835	0.0003	0.1%	6.5%

Environmental vs Technical (bootstrapped):

Metric	Tech – Env	95% CI	Sig.?
ROC-AUC	-6.4pp	[-7.8, -5.1]	Yes
PR-AUC	-2.8pp	[-3.3, -2.2]	Yes
Precision	-6.7pp	[-7.9, -5.5]	Yes
Recall	+1.8pp	[-0.04, +3.7]	No

The model predicts environmental outages significantly better than technical ones (6.4pp ROC-AUC gap, CI excludes zero).

Why this makes scientific sense:

- **Environmental outages are weather-driven.**
The model uses weather features *and* weather-correlated NTL dimming. Both channels carry information.
- **Technical outages** are driven by equipment state, human factors, random failures — no observable signature in NTL or weather.
- **Planned outages** show high ROC-AUC (0.889) but zero useful precision — they happen in generally “risky” municipalities but the model cannot predict *when*.

► **Recall difference is not significant** — the model catches similar fractions, but technical “catches” are largely accidental.

Direct Comparison: Binary vs LTR

Benchmark table — same test period, alerts/day $\approx$ 10:

Model	Method	Precision	Recall	Alerts/day	Calibration
Binary	Platt	12.5%	9.7%	7.6	Log-loss 0.021
Binary	Beta	12.2%	10.7%	8.7	Log-loss 0.021
Binary	Raw	9.9%	16.2%	16.1	Log-loss 0.300
LTR	rank:pairwise	10.2%	10.4%	10.0 (fixed)	N/A

Key findings: (1) At $\sim$ 10 alerts/day, both achieve $\sim$ 10% recall — neither dominates. (2) Calibrated binary has **higher precision** (12.5% vs 10.2%) because it's more selective. (3) LTR forces exactly 10 alerts/day even on quiet days. (4) **Calibration is the differentiator** — LTR scores cannot serve as probabilities.

Source: `benchmark_table_same_run.csv` — binary run `20260223_125430` vs LTR run `20260220_114434`.

When Each Approach Wins

Scenario	Better approach	Why
“Inspect exactly 10 municipalities/day”	LTR	Directly optimises within-day ranking
“Alert when risk > threshold; adapt daily”	Binary + calibration	Flexible, adapts to daily risk
“Communicate risk to stakeholders”	Binary + calibration	“5% chance” is interpretable
“Feed into severity/count model”	Binary + calibration	P(outage) is a meaningful input
“Benchmark under fixed budget”	Both	LTR provides the fixed-K reference

Conclusion: The calibrated binary classifier is the **primary model** — more general, interpretable, and composable. LTR is a **secondary benchmark** that validates performance under a specific fixed-K constraint. Both confirm the NTL + weather signal.

What the Model Can and Cannot Do

Can do:

- ▶ **Calibrated outage probabilities** using only free satellite + weather data
- ▶ **Strong discrimination** (ROC-AUC 0.879) — no grid telemetry required
- ▶ **Concentrate risk** — 12.5% precision at 8 alerts/day (25× base rate)
- ▶ Predict **environmental outages better** than technical (confirmed statistically)
- ▶ **Ranking transfers** across unseen regions (98.5% ROC-AUC)
- ▶ **Honest probabilities** after Platt (ECE $\approx$ 0.001)
- ▶ **Flexible alerting** via policy-based thresholds

Cannot do:

- ▶ Achieve high precision at low volumes (best $\sim$ 12.5%)
- ▶ Predict **technical outages** as well as environmental
- ▶ **Transfer calibration** to unseen regions without re-calibration
- ▶ Detect planned maintenance or crime-related outages
- ▶ Replace real-time grid monitoring (SCADA)

Structure:

1. Calibrated probabilistic forecasting (primary)
2. NTL + weather feature story (ablation)
3. Calibration as essential step (Platt)
4. Environmental > Technical (cause analysis)
5. Temporal stability + spatial generalization
6. LTR as secondary comparator

Next Steps

Immediate:

- ▶ **Rebuild panel** with new label columns (total duration, cause counts, mixed-cause flag) — code ready, awaiting NTL data rebuild
- ▶ **Mixed-cause sensitivity** for script 13 (expected minimal impact: 1.7% ambiguity)
- ▶ Re-run cause-specific eval on updated panel

Extensions:

- ▶ **Count modeling** ($n_outages$ | outage occurred): Hurdle model — Poisson/NB on positive nights only
- ▶ **Duration modeling** ($total_length_min$ | outage): Log-normal regression on positives
- ▶ **Decomposed expected burden:**
$$E[disruption] = P(outage) \times E[duration \mid outage]$$
- ▶ Lock tables/figures for chapters

Summary

Metric	Value
Model	XGBoost + Platt calibration
Task	Forecast (strict t-1)
Features	82 (66 in forecast mode)
ROC-AUC	0.879
PR-AUC	0.056 (14× random)
Precision	12.5% at 8 alerts/day
Base rate	0.40%
Calibration	ECE ≈ 0.001 (Platt)
Env vs Tech	ROC-AUC gap -6.4pp (sig.)
Spatial holdout	98.5% ROC-AUC retention

Key contributions:

1. **Calibrated probabilistic forecast** from NTL data alone — no grid telemetry needed
2. **NTL + weather** is the dominant feature combination (systematic ablation)
3. **Post-hoc calibration is essential** — raw XGBoost probabilities are useless
4. **Environmental outages are significantly more predictable** — confirms NTL+weather signal pathway
5. **Flexible policy-based alerting** replaces fixed top-K decision rule
6. **LTR benchmark** validates under fixed-budget assumptions

Thank You

Key scripts:

- ▶ `ML_Mexico_final.R` — NTL pipeline + panel building
- ▶ `scripts/05b_binary_threshold_eval_main.R` — Main eval
- ▶ `scripts/07b_probcal_ablations.R` — Ablation orchestrator
- ▶ `scripts/13_cause_specific_eval.R` — Cause analysis

Key outputs:

- ▶ `data/baselines/binary_threshold/20260223_125430_forecast_strict/`
- ▶ Ablation batch `20260302_122455`
- ▶ `benchmark_table_same_run.csv` (binary vs LTR)

Data sources:

Contact:

armande.aboudrar-meda@cmcc.it

Discussion:

- ▶ How should we frame the probability vs ranking comparison?
- ▶ Which operational scenario to prioritise for deployment framing?
- ▶ Count/duration model: prioritise Poisson hurdle or log-normal severity?

- ▶ NASA Black Marble (VIIRS VNP46A2)
- ▶ EU JRC GHS-BUILT-S
- ▶ ECMWF ERA5